

UNIVERSITY OF LJUBLJANA

Faculty of Computer and Information Science
Laboratory for Data Technologies
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RESEARCH INTERESTS

NETWORK SCIENCE | PRIMARY RESEARCH AREA

Structure and dynamics of complex social, information & technological networks
Methods and algorithms for network modeling, sampling & comparison, clustering & partitioning, link analysis
Applications in bibliometrics & scientometrics, fraud & anomaly detection, software engineering

GRAPH MACHINE LEARNING | SECONDARY RESEARCH AREA

Machine learning with graphs, graph representation learning & relational deep learning

DATA SCIENCE | OTHER FIELDS OF INTEREST

Selected areas of data analysis, data mining, stream mining, statistics

EDUCATION

PhD | COMPUTER AND INFORMATION SCIENCE

University of Ljubljana | Faculty of Computer and Information Science
Thesis "Detecting groups of nodes in networks using label propagation" | Advisor **Marko Bajec**
Course grade average: 9.82 / 10.0 | Thesis grade: 10
Oct 2008 – June 2013

DIPLOMA [BSc] | COMPUTER SCIENCE AND MATHEMATICS

University of Ljubljana | Faculty of Computer and Information Science & Faculty of Mathematics and Physics
Diploma "Detecting fraud based on the analysis of social networks" | Advisors **Marko Bajec** & **Matjaž Kukar** | **Prešeren award**
Course grade average: 9.78 / 10.0 | Diploma grade: 10 | Summa Cum Laude
Oct 2004 – Sep 2008

POSITIONS

ASSOCIATE PROFESSOR | COMPUTER AND INFORMATION SCIENCE

University of Ljubljana | Faculty of Computer and Information Science
Oct 2022 – Oct 2027

RESEARCH FELLOW | METHODOLOGY, STATISTICS AND INFORMATICS

University of Ljubljana | Faculty of Social Sciences
Jan 2018 – Dec 2026

TEACHER & ASSISTANT

University of Ljubljana | Faculty of Computer and Information Science
Oct 2015 – & July – Dec 2013

[POSTDOCTORAL] RESEARCHER

University of Ljubljana | Faculty of Computer and Information Science
Oct 2008 – June 2015

FUNDING

Slovenian Ministry of Education, Science and Sport & European Union, European Social Fund
Development of algorithms for network analysis in large company | Funding awarded **73,152 €**
Jan 2014 – June 2015

AWARDS & HONORS

- 2017 **UNIVERSITY AWARD** | University of Ljubljana | Young university professors
- 2012 **NATIONAL AWARD** | Slovenian Research Agency | Journal publication
- 2011 **FACULTY AWARD** | University of Ljubljana | Faculty of Computer and Information Science | Journal publication
- 2008 **PREŠEREN AWARD** | University of Ljubljana | Faculty of Computer and Information Science | BSc thesis
- 2008 **1ST PLACE** | University of California, San Diego | Data Mining Contest | Undergraduate division
- 2006 **RECTOR'S LIST** | University of Ljubljana | Top university students
- 2005 – 08 **DEAN'S LIST** | University of Ljubljana | Faculty of Computer and Information Science | Top faculty students
- 2000 – 08 **ZOIS FELLOWSHIP** | Slovene Human Resources and Scholarship Fund | Top national students

TALKS & SEMINARS

- “Intermediacy and convexity in networks” | Department seminar (**Central European University**, Department of Network and Data Science, 2025) | Host **Márton Pósfai**
- “Network abstraction with backbones and skeletons: Spanning trees versus convex skeletons” | Department seminar (**Imperial College London**, Centre for Complexity Science, 2022) | Host **Tim Evans**
- “On computing spanning trees of unweighted networks” | *Wednesday Seminar* (University of Ljubljana, Faculty of Mathematics and Physics, 2021) | Host **Vladimir Batagelj**
- “Intermediacy of publications: Uncovering important publications for the development of a field” | At: *6th European Conference on Statistical Network Science COSTNET* (online, 2020)
- “Label propagation for clustering” | At: *6th Slovenian Conference on Network Science NetSlo* (Ljubljana, Slovenia, 2020)
- “Intermediacy of publications” | At: *5th Slovenian Conference on Network Science NetSlo* (Ljubljana, Slovenia, 2019)
- “Intermediacy of publications” | *AI Seminar* (University of Ljubljana, Faculty of Computer and Information Science, 2019) | Host **Marko Robnik-Šikonja**
- “Convexity in complex networks” | *LCN2 Seminar* (**Leiden University**, Institute of Advanced Computer Science, 2017) | Host **Frank W. Takes**
- “Network analysis in bibliometrics” | Department seminar (**Leiden University**, Centre for Science and Technology Studies, 2017) | Host **Vincent Traag**
- “On convexity in complex networks” | At: *6th International Workshop on Social Network Analysis ARS* (Naples, Italy, 2017)
- “Should you (not) publish in computer science journals?” | *Cookie Seminar* (University of Ljubljana, Faculty of Computer and Information Science, 2017) | Host **Sebastijan Šprager**
- “Comparison of methods for clustering scientific publications based on citations” | *IBMI Seminar* (University of Ljubljana, Faculty of Medicine, 2017) | Host **Nataša Kejžar**
- “Reliability of bibliographic databases for scientometrics network analysis” | At: *8th International Conference on Information Technologies and Information Society ITIS* (Šmarješke toplice, Slovenia, 2016)
- “On convexity in complex networks” | *Wednesday Seminar* (University of Ljubljana, Faculty of Mathematics and Physics, 2016) | Host **Vladimir Batagelj**
- “Network analysis vs WikiLeaks” | *Cookie Seminar* (University of Ljubljana, Faculty of Computer and Information Science, 2015) | Host **Janez Demšar**
- “Selected topics from network analysis” | *Wednesday Seminar* (University of Ljubljana, Faculty of Mathematics and Physics, 2014) | Host **Vladimir Batagelj**
- “Network analysis” | Habilitation lecture (University of Ljubljana, Faculty of Computer and Information Science, 2014) | Host **Igor Kononenko**
- “Network and link analysis” | Department seminar (**University of West Bohemia**, Department of Computer Science and Engineering, 2014) | Host **Dalibor Fiala**
- “Detecting groups of nodes in networks by label propagation” | Invited lecture (Faculty of Information Studies in Novo mesto, 2014) | Host **Zoran Levnajic**
- “An expert system for detecting automobile insurance fraud using social network analysis” | Invited talk (Slovenian Academy of Sciences and Arts, 2013) | **National award**
- “Detecting insurance fraud using network analysis” | *Seminar for Probability, Statistics and Financial Mathematics* (University of Ljubljana, Faculty of Mathematics and Physics, 2013) | Host **Aleš Toman**
- “Structured-world conjecture: On modules and communities in real-world networks” | Group seminar (**Stanford University**, Computer Science Department, 2012) | Host **Jure Leskovec**
- “Structured-world conjecture: On modules and communities in real-world networks” | *AI Seminar* (**University of California, San Diego**, Department of Computer Science and Engineering, 2012) | Host **Charles Elkan**
- “Introduction to data networks analysis” | Invited lecture (University of Ljubljana, Faculty of Computer and Information Science, 2010) | Host **Marko Bajec**

PUBLICATIONS & PATENTS

JOURNAL PUBLICATIONS

- Lovro Šubelj | “Computing well-balanced spanning trees of unweighted networks” | **Algorithms** 18(12), 760 (2025) | **Journal cover**
- Romi Koželj, Lovro Šubelj & Jurij Bajc | “Methods for representing earthquake time series with networks” | *Informatica* 48(14), 1-26 (2024)
- Žiga Babnik, Jasmina Pegan, Domen Kos & Lovro Šubelj | “Generating lyrics using constrained random walks on a word network” | *Informatica* 48(1), 131-140 (2024)
- Milan Jović, Lovro Šubelj, Tea Golob, Matej Makarovič, Taha Yasseri, Danijela Boberić Krstićev, Srdjan Škrbić & Zoran Levnajić | “Terrorist attacks sharpen the binary perception of “Us” vs. “Them”” | **Scientific Reports** 13, 12451 (2023)
- Vincent Traag & Lovro Šubelj | “Large network community detection by fast label propagation” | **Scientific Reports** 13, 2701 (2023)
- Tim Poštuvan, Semir Salkić & Lovro Šubelj | “Learning-based link prediction analysis for Facebook100 network” | *Uporabna Informatika* 29(2), 83-94 (2021)
- Ilya Makarov, Dmitry Kiselev, Nikita Nikitinsky & Lovro Šubelj | “Survey on graph embeddings and their applications to machine learning problems on graphs” | *PeerJ Computer Science* 7, e357 (2021)
- Anej Svete, Jakob Hostnik & Lovro Šubelj | “It is not just about the melody: How Europe votes for its favorite songs” | *Uporabna Informatika* 28(2), 66-84 (2020)
- Jaka Stavanja, Matej Klemen & Lovro Šubelj | “Predicting kills in Game of Thrones using network analysis” | *Uporabna Informatika* 28(2), 55-65 (2020)
- Lovro Šubelj, Ludo Waltman, Vincent Traag & Nees Jan van Eck | “Intermediacy of publications” | **Royal Society Open Science** 7(1), 190207 (2020)
- Luka Naglič & Lovro Šubelj | “War pact model of shrinking networks” | **PLoS ONE** 14(10), e0223480 (2019)
- Lovro Šubelj, Dalibor Fiala, Tadej Ciglarič & Luka Kronegger | “Convexity in scientific collaboration networks” | **Journal of Informetrics** 13(1), 10-31 (2019)
- Vitjan Zavrtanik & Lovro Šubelj | “Community detection in Slovene public spending” | *Uporabna Informatika* 26(3), 107-117 (2018)
- Jaka Šircelj, Romi Koželj & Lovro Šubelj | “Effects of species extinction on ecosystems stability” | *Uporabna Informatika* 26(3), 118-127 (2018)
- Lovro Šubelj | “Convex skeletons of complex networks” | **Journal of the Royal Society Interface** 15(145), 20180422 (2018)
- Tilen Marc & Lovro Šubelj | “Convexity in complex networks” | **Network Science** 6(2), 176-203 (2018)
- Neli Blagus, Lovro Šubelj & Marko Bajec | “Empirical comparison of network sampling: How to choose the most appropriate method?” | **Physica A** 477, 136-148 (2017)
- Lovro Šubelj & Dalibor Fiala | “Publication boost in Web of Science journals and its effect on citation distributions” | **JASIST** 68(4), 1018-1023 (2017)
- Lovro Šubelj, Nees Jan van Eck & Ludo Waltman | “Clustering scientific publications based on citation relations: A systematic comparison of different methods” | **PLoS ONE** 11(4), e0154404 (2016)
- Lovro Šubelj, Marko Bajec, Biljana M. Boshkoska, Andrej Kastrin & Zoran Levnajić | “Quantifying the consistency of scientific databases” | **PLoS ONE** 10(5), e0127390 (2015)
- Neli Blagus, Lovro Šubelj, Gregor Weiss & Marko Bajec | “Sampling promotes community structure in social and information networks” | **Physica A** 432, 206-215 (2015)
- Dalibor Fiala, Lovro Šubelj, Slavko Žitnik & Marko Bajec | “Do PageRank-based author rankings outperform simple citation counts?” | **Journal of Informetrics** 9(2), 334-348 (2015)
- Lovro Šubelj, Dalibor Fiala & Marko Bajec | “Network-based statistical comparison of citation topology of bibliographic databases” | **Scientific Reports** 4, 6496 (2014)
- Lovro Šubelj, Slavko Žitnik, Neli Blagus & Marko Bajec | “Node mixing and group structure of complex software networks” | **Advances in Complex Systems** 17(7), 1450022 (2014)
- Neli Blagus, Lovro Šubelj, Aljaž Zrnc, Marko Janković & Marko Bajec | “Preserving properties under simplification of social networks” | *Uporabna Informatika* 22(3), 127-136 (2014) | In Slovene
- Neli Blagus, Lovro Šubelj & Marko Bajec | “Assessing the effectiveness of real-world network simplification” | **Physica A** 413, 134-146 (2014)
- Slavko Žitnik, Lovro Šubelj & Marko Bajec | “SkipCor: Skip-mention coreference resolution using linear-chain conditional random fields” | **PLoS ONE** 9(6), e100101 (2014)
- Marko Janković, Slavko Žitnik, Lovro Šubelj, Neli Blagus, Aljaž Zrnc & Marko Bajec | “Improvement of software development method with minimal involvement of developers” | *Uporabna Informatika* 22(1), 3-11 (2014) | In Slovene
- Lovro Šubelj & Marko Bajec | “Group detection in complex networks: An algorithm and comparison of the state of the art” | **Physica A** 397, 144-156 (2014)
- Lovro Šubelj, Slavko Žitnik, Marko Janković, Bojan Klemenc, Aleš Kumer, Aljaž Zrnc & Marko Bajec | “Large real-world networks” | *Uporabna Informatika* 21(2), 107-115 (2013) | In Slovene

- Slavko Žitnik, **Lovro Šubelj**, Dejan Lavbič, Olegas Vasilecas & Marko Bajec | “General context-aware data matching and merging framework” | *Informatica* 24(1), 119-152 (2013)
- Dejan Lavbič, Slavko Žitnik, **Lovro Šubelj**, Aleš Kumer, Aljaž Zrnc & Marko Bajec | “Traversal and relations discovery among business entities and people using semantic web technologies and trust management” | *Databases and Information Systems* 7, 164-174 (2013)
- Aljaž Zrnc, Dejan Lavbič, **Lovro Šubelj**, Slavko Žitnik, Aleš Kumer & Marko Bajec | “NoSQL databases” | *Uporabna Informatika* 20(3), 164-172 (2012) | In Slovene
- **Lovro Šubelj** & Marko Bajec | “Ubiquitousness of link-density and link-pattern communities in real-world networks” | *European Physical Journal B* 85(1), 32 (2012) | **Editors' highlight**
- Neli Blagus, **Lovro Šubelj** & Marko Bajec | “Self-similar scaling of density in complex real-world networks” | *Physica A* 391(8), 2794-2802 (2012)
- **Lovro Šubelj** & Marko Bajec | “Robust network community detection using balanced propagation” | *European Physical Journal B* 81(3), 353-362 (2011)
- **Lovro Šubelj** & Marko Bajec | “Community structure of complex software systems: Analysis and applications” | *Physica A* 390(16), 2968-2975 (2011)
- **Lovro Šubelj** & Marko Bajec | “Unfolding communities in large complex networks: Combining defensive and offensive label propagation for core extraction” | *Physical Review E* 83(3), 036103 (2011) | **Faculty award**
- **Lovro Šubelj**, Štefan Furlan & Marko Bajec | “An expert system for detecting automobile insurance fraud using social network analysis” | *Expert Systems with Applications* 38(1), 1039-1052 (2011) | **National award**
- Bojan Klemenc, Peter Ciuha, **Lovro Šubelj** & Marko Bajec | “Visual and aural: Visualization of harmony in music with color” | *IPSI BgD Transactions on Internet Research* 7(1), 48-53 (2011)
- **Lovro Šubelj**, David Jelenc, Eva Zupančič, Dejan Lavbič, Denis Trček, Marjan Krisper & Marko Bajec | “Merging data sources based on semantics, contexts and trust” | *IPSI BgD Transactions on Internet Research* 7(1), 18-30 (2011)

PUBLICATION PREPRINTS

- **Lovro Šubelj** & Marko Bajec | “Clustering assortativity, communities and functional modules in real-world networks” | e-print *arXiv:1202.3188v1*, pp. 21

CONFERENCE PUBLICATIONS

- Vid Stropnik & **Lovro Šubelj** | “Predicting successful passes in football: Benchmark dataset and GNN ablation study” | In: *Proceedings of the 20th International Conference on Network Science NetSci* (Maastricht, Netherlands, 2025), p. 1
- Vincent Traag & **Lovro Šubelj** | “Fast label propagation algorithm for community detection” | In: *Proceedings of the 18th International Conference on Network Science NetSci* (Vienna, Austria, 2023), p. 1
- Nejc Rožman, Marko Corn, Gašper Škulj, Janez Diaci & **Lovro Šubelj** | “Emergence of a scale-free network topology in a blockchain-based shared manufacturing” | In: *Proceedings of the 3rd International Conference on Blockchain Computing and Applications BCAA* (Tartu, Estonia, 2021), pp. 172-178
- **Lovro Šubelj** | “Spanning trees that preserve network distances” | In: *Proceedings of the 1st Joint Sunbelt and NetSci Conference Networks* (Washington D.C., USA, 2021), pp. 270-271
- Luka Naglič & **Lovro Šubelj** | “War pact network model: A generative model of networks that shrink” | In: *Proceedings of the 4th European Social Networks Conference EUSN* (Zurich, Switzerland, 2019), p. 240
- **Lovro Šubelj**, Ludo Waltman, Vincent Traag & Nees Jan van Eck | “Intermediacy of publications” | In: *Proceedings of the 17th International Conference on Informetrics and Scientometrics ISSI* (Rome, Italy, 2019), pp. 1288-1300
- **Lovro Šubelj**, Ludo Waltman, Vincent Traag & Nees Jan van Eck | “Intermediacy of publications” | In: *Proceedings of the 5th International Conference on Computational Social Science IC²S²* (Amsterdam, Netherlands, 2019), pp. 1-3
- **Lovro Šubelj**, Ludo Waltman, Vincent Traag & Nees Jan van Eck | “Uncovering important intermediate publications” | In: *Proceedings of the 38th INSNA International Social Network Conference Sunbelt* (Utrecht, Netherlands, 2018), p. 1
- **Lovro Šubelj** | “Convex skeleton: Generalization of a network spanning tree” | In: *Proceedings of the 1st NetSci Workshop on Large and Rich Network Structures LaRichNet* (Paris, France, 2018), p. 1
- **Lovro Šubelj**, Ludo Waltman, Vincent Traag & Nees Jan van Eck | “Intermediacy of publications” | In: *Proceedings of the 13th International Conference on Network Science NetSci* (Paris, France, 2018), p. 1
- Tilen Marc & **Lovro Šubelj** | “Three forms of convexity in graphs and networks” | In: *Proceedings of the 2nd European Conference on Statistical Network Science COSTNET* (Palma de Mallorca, Spain, 2017), p. 54
- Marko Mišić, Dražen Drašković, **Lovro Šubelj** & Marko Bajec | “Parallel implementation of the label propagation method for community detection on the GPU” | In: *Proceedings of the 26th International Electrotechnical and Computer Science Conference ERK* (Portorož, Slovenia, 2017), pp. 359-362
- **Lovro Šubelj** & Tilen Marc | “On convexity in complex networks” | In: *Proceedings of the 8th International Conference on Complex Networks CompleNet* (Dubrovnik, Croatia, 2017), pp. 14-16
- **Lovro Šubelj**, Nees Jan van Eck & Ludo Waltman | “Comparison of methods for clustering citation networks” | In: *Proceedings of the 2nd International Conference on Network Science X NetSci-X* (Wrocław, Poland, 2016), p. 1
- **Lovro Šubelj** | “Exploratory and predictive tasks of network community detection” | In: *Proceedings of the 10th International Conference on Network Science NetSci* (Zaragoza, Spain, 2015), p. 1

- **Lovro Šubelj**, Dalibor Fiala & Marko Bajec | “Consistency of citation topology of bibliographic databases” | In: *Proceedings of the 10th International Conference on Network Science NetSci* (Zaragoza, Spain, 2015), p. 1
- **Lovro Šubelj** | “Large network community detection in practical scenarios” | In: *Proceedings of the 5th International Workshop on Social Network Analysis ARS* (Capri, Italy, 2015), p. 78
- **Lovro Šubelj** & Marko Bajec | “Network group discovery by hierarchical label propagation” | In: *Proceedings of the 1st European Social Networks Conference EUSN* (Barcelona, Spain, 2014), p. 284
- **Lovro Šubelj**, Zoran Bosnić, Matjaž Kukar & Marko Bajec | “Automatization of the stream mining process” | In: *Proceedings of the 26th International Conference on Advanced Information Systems CAISE* (Thessaloniki, Greece, 2014), pp. 409-423
- **Lovro Šubelj**, Slavko Žitnik & Marko Bajec | “Who reads and who cites? Unveiling author citation dynamics by modeling citation networks” | In: *Proceedings of the 9th International Conference on Network Science NetSci* (Berkeley, California, USA, 2014), p. 1
- **Lovro Šubelj**, Gregor Weiss, Neli Blagus & Marko Bajec | “What coins the bitcoin? Exploratory analysis of bitcoin market value by network group discovery” | In: *Proceedings of the 9th International Conference on Network Science NetSci* (Berkeley, California, USA, 2014), p. 1
- Neli Blagus, **Lovro Šubelj**, Gregor Weiss & Marko Bajec | “Large networks grow smaller: How to choose the right simplification method?” | In: *Proceedings of the 9th International Conference on Network Science NetSci* (Berkeley, California, USA, 2014), p. 1
- Slavko Žitnik, **Lovro Šubelj**, Marko Janković, Bojan Furlan, Dražen Drašković, Nemanja Kojić, Marko Mišić & Marko Bajec | “Iterative end-to-end information extraction based on linear models” | In: *Proceedings of the 22nd International Electrotechnical and Computer Science Conference ERK* (Portorož, Slovenia, 2013), pp. 47-50
- Miloš Pavković, Dražen Drašković, **Lovro Šubelj**, Slavko Žitnik, Dejan Lavbič, Marko Janković, Jelica Protić & Boško Nikolić | “Intelligent techniques for searching Internet forums” | In: *Proceedings of the 22nd International Electrotechnical and Computer Science Conference ERK* (Portorož, Slovenia, 2013), pp. 85-88
- **Lovro Šubelj**, Marko Bajec & Neli Blagus | “Group extraction for real-world networks: The case of communities, modules, and hubs and spokes” | In: *Proceedings of the 8th International Conference on Network Science NetSci* (Copenhagen, Denmark, 2013), pp. 152-153
- Neli Blagus, Marko Bajec & **Lovro Šubelj** | “Preserving properties under simplification of real-world networks” | In: *Proceedings of the 8th International Conference on Network Science NetSci* (Copenhagen, Denmark, 2013), p. 166
- **Lovro Šubelj** & Marko Bajec | “Model of complex networks based on citation dynamics” | In: *Proceedings of the 2nd WWW Workshop on Large Scale Network Analysis LSNA* (Rio de Janeiro, Brazil, 2013), pp. 527-530
- Slavko Žitnik, **Lovro Šubelj**, Dejan Lavbič, Aljaž Zrnec & Marko Bajec | “Collective information extraction using first-order probabilistic models” | In: *Proceedings of the 5th Balkan Conference in Informatics BCI* (Novi Sad, Serbia, 2012), pp. 279-282
- **Lovro Šubelj** & Marko Bajec | “Software systems through complex networks science: Review, analysis and applications” | In: *Proceedings of the 1st KDD Workshop on Software Mining SoftMine* (Beijing, China, 2012), pp. 9-16
- Dejan Lavbič, Slavko Žitnik, **Lovro Šubelj**, Aleš Kumer, Aljaž Zrnec & Marko Bajec | “Traversal and relations discovery among business entities and people using semantic web technologies and trust management” | In: *Proceedings of the 10th International Baltic Conference on Databases and Information Systems DB&IS* (Vilnius, Lithuania, 2012), pp. 11
- **Lovro Šubelj**, Slavko Žitnik, Aljaž Zrnec, Aleš Kumer, Bojan Klemenc, Dejan Lavbič & Marko Bajec | “Properties of large real-world networks and some applications” | In: *Zbornik 19. konference Dnevi slovenske informatike DSI* (Portorož, Slovenia, 2012), pp. 10 | In Slovene
- Aljaž Zrnec, Dejan Lavbič, **Lovro Šubelj**, Slavko Žitnik, Aleš Kumer & Marko Bajec | “NoSQL databases” | In: *Zbornik 19. konference Dnevi slovenske informatike DSI* (Portorož, Slovenia, 2012), pp. 10 | In Slovene
- Marko Bajec, Štefan Furlan, Aleš Kumer, **Lovro Šubelj**, Slavko Žitnik & Dejan Lavbič | “Supervisor+” | In: *Zbornik 3. konference Informatika v javni upravi IJU* (Brdo pri Kranju, Slovenia, 2011), pp. 6 | In Slovene
- **Lovro Šubelj** & Marko Bajec | “Generalized network community detection” | In: *Proceedings of the 1st ECML PKDD Workshop on Finding Patterns of Human Behaviors in Network and Mobility Data NEMO* (Athens, Greece, 2011), pp. 66-84
- **Lovro Šubelj**, Neli Blagus, Štefan Furlan, Bojan Klemenc, Aleš Kumer, Dejan Lavbič, Aljaž Zrnec, Slavko Žitnik & Marko Bajec | “Complex networks analysis: Introduction and applications” | In: *Zbornik 2. konference Informatika v javni upravi IJU* (Brdo pri Kranju, Slovenia, 2010), pp. 10 | In Slovene
- Slavko Žitnik, Bojan Klemenc, Dejan Lavbič, **Lovro Šubelj**, Štefan Furlan, Aleš Kumer, Aljaž Zrnec, Neli Blagus & Marko Bajec | “Trends in intelligent data retrieval from heterogeneous data sources” | In: *Zbornik 2. konference Informatika v javni upravi IJU* (Brdo pri Kranju, Slovenia, 2010), pp. 12 | In Slovene
- **Lovro Šubelj** & Marko Bajec | “Unfolding network communities by combining defensive and offensive label propagation” | In: *Proceedings of the 1st ECML PKDD Workshop on the Analysis of Complex Networks ACNE* (Barcelona, Spain, 2010), pp. 87-104
- Bojan Klemenc, Peter Ciuha, **Lovro Šubelj** & Marko Bajec | “Visual and aural: Visualization of harmony in music with color” | In: *Proceedings of the Symposia on Internet Related Research VIPSI* (Tivat, Montenegro, 2010), pp. 6
- **Lovro Šubelj**, David Jelenc, Eva Zupančič, Dejan Lavbič, Denis Trček, Marjan Krisper & Marko Bajec | “Merging data sources based on semantics, contexts and trust” | In: *Proceedings of the Symposia on Internet Related Research VIPSI* (Amalfi, Italy, 2010), pp. 13

- Lovro Šubelj, Štefan Furlan & Marko Bajec | “Detecting fraud based on the analyses of social networks” | In: *Zbornik 16. konference Dnevi slovenske informatike DSI* (Portorož, Slovenia, 2009), pp. 10 | In Slovene
- Giovanni Godena, Janez Tancek, Igor Steiner, David Čuk, Lovro Šubelj, Jože Grilj & Peter Kosin | “LiteBatch: A tool for batch process control on the PLC platform” | In: *Zbornik 5. konference Avtomatizacija v industriji in gospodarstvu AIG* (Maribor, Slovenia, 2007), pp. 51-56 | In Slovene

BOOK CHAPTERS

- Lovro Šubelj | “Label propagation for clustering” | In: *Advances in Network Clustering and Blockmodeling* (Wiley, New York, 2020), pp. 121-150

PRESS COVERAGE

- Mojca Vizjak Pavšič | “Zakaj je Drnovšek premagal Bulca” | *Delo* (February 21, 2019) | In Slovene
- Mojca Vizjak Pavšič | “NetSlo’17 o znanosti 21. stoletja” | *Delo* (February 9, 2017) | In Slovene
- Lovro Šubelj & Jernej Ule | “Bruci, vstopate v razburljivi del življenja. Lp, vaš profesor” | *Delo* (October 1, 2014) | In Slovene
- Uroš Škerl Kramberger | “George ne bo shodil brez države” | *Dnevnik* (March 28, 2013) | In Slovene

PATENTS & APPLICATIONS

- Lovro Šubelj & Marko Bajec | “The process for allocation of computational resources in networks of distributed devices” | *Slovenian Institute of Intellectual Property SI-24553-A* (Ljubljana, Slovenia, 2015), P-201300392, pp. 5 | In Slovene
- Lovro Šubelj, Slavko Žitnik, Dejan Lavbič, Aleš Kumer & Marko Bajec | “Combining heterogeneous data sources of data processing devices” | *Slovenian Institute of Intellectual Property* (Ljubljana, Slovenia, 2013), P-201300263, pp. 7 | In Slovene

SERVICE TO PROFESSION

SCIENTIFIC AFFILIATION

- 2020 – Slovenian Association for Computing Machinery SACM
- 2020 – Association for Computing Machinery ACM
- 2015 – Slovenian Network Science Society **NetSlo**
- 2014 – Network Science Society **NetSci**
- 2010 – Slovenian Informatics Society SDI
- 2009 – Slovenian Artificial Intelligence Society SLAIS

JOURNAL EDITORIAL

- 2021 – **Entropy** | Topic editor
- 2017 – **PLoS ONE** | Academic editor
- 2015 – *Journal of Computer Science and Engineering* | Editorial board

JOURNAL ADVISORY

- 2024 *Journal of Computational Science* (Elsevier) | Reviewer
- 2016 | 24 **PLoS ONE** (Public Library of Science) | Regular reviewer
- 2023 *Journal of Classification* (Springer) | Reviewer
- 2019 | 23 **Quantitative Science Studies** (MIT Press) | Regular reviewer
- 2014 | 20 | 22 **Scientific Reports** (Nature Publishing Group) | Regular reviewer
- 2014 | 21 **Social Network Analysis and Mining** (Springer) | Regular reviewer
- 2020 **Journal of the Royal Society Interface** (Royal Society Publishing) | Reviewer
- 2016 | 17 | 20 **Scientometrics** (Springer) | Regular reviewer
- 2016 | 17 | 20 **Journal of Informetrics** (Elsevier) | Regular reviewer
- 2015 | 18 | 20 **Entropy** (Multidisciplinary Digital Publishing Institute) | Regular reviewer
- 2017 | 19 **Journal of Complex Networks** (Oxford University Press) | Regular reviewer
- 2018 **IEEE Access** (IEEE) | Reviewer
- 2012 – 18 **Physica A** (Elsevier) | Regular reviewer
- 2017 **European Physical Journal B** (Springer) | Reviewer
- 2017 *Italian Journal of Applied Statistics* (Padua) | Reviewer
- 2017 *Journal of Computer Science and Engineering* (Peertechz) | Reviewer
- 2017 *Communications in Mathematical and in Computer Chemistry* (MATCH) | Reviewer
- 2016 **Social Networks** (Elsevier) | Reviewer

2016 *Expert Systems with Applications* (Elsevier) | Reviewer
 2016 *IEEE Transactions on Neural Networks and Learning Systems* (IEEE) | Reviewer
 2015 *IEEE Transactions on Cybernetics* (IEEE) | Reviewer
 2015 *Neural Computing and Applications* (Springer) | Reviewer
 2014 *Journal of Systems and Software* (Elsevier) | Reviewer
 2014 *Modern Physics Letters B* (World Scientific) | Reviewer
 2013 | 14 *International Journal of Modern Physics C* (World Scientific) | Regular reviewer
 2013 *Advances in Complex Systems* (World Scientific) | Reviewer
 2013 *Knowledge-Based Systems* (Elsevier) | Reviewer
 2012 *Chinese Physics B* (IOP Publishing) | Reviewer
 2011 – 15 *Journal of Computer Science and Technology* (Springer) | Regular reviewer
 2010 *International Journal of Information System Modeling and Design* (IGI Publishing) | Reviewer
 2010 | 19 – 21 *Uporabna Informatika* (SDI) | Regular reviewer

CONFERENCE COMMITTEE

2022 – 24 | 26 *ECML PKDD Workshop on Mining and Learning With Graphs* **MLG** | Program committee
 2025 | 26 *International Conference on Network Science* **NetSci** | Program committee
 2017 – 26 *Slovenian Conference on Network Science* **NetSlo** | Organizing committee | With many others
 2022 | 25 *Student Computing Research Symposium* **SCORES** | Organizing committee | With many others
 2023 – 25 *International Conference on Web Search and Data Mining* **WSDM** | Program committee
 2023 | 24 *International Conference on Complex Networks and their Applications* **CNA** | Program committee
 2022 | 23 *The Web Conference* **Web** | Program committee
 2021 *International Conference on Modeling and Analysis of Complex Systems and Processes* **MACSPro** | Program committee
 2021 *The Web Conference* **Web** | Local volunteers chair
 2015 | 16 *Networking Slovenian Networkers* **MSN** | Organizing committee | With many others
 2014 – 19 *International Conference on Information and Software Technologies* **ICIST** | Program committee
 2014 – 18 *International Conference on Research Challenges in Information Science* **RCIS** | Program & poster committee
 2014 – 18 *International Conference on Information Technologies and Information Society* **ITIS** | Program committee
 2012 | 18 *International Baltic Conference on Databases and Information Systems* **DB&IS** | Program committee
 2017 *International Electrotechnical and Computer Science Conference* **ERK** | Program committee
 2009 – 11 | 15 *International Conference on Advanced Information Systems* **CAISE** | Additional reviewer

CONFERENCE ATTENDANCE

2014 | 15 | 18 | 22 | 23 | 25 *International Conference on Network Science* **NetSci**
 2017 – 20 | 23 | 24 *Slovenian Conference on Network Science* **NetSlo**
 2014 | 19 | 23 *European Social Networks Conference* **EUSN**
 2022 | 23 *Learning on Graphs Conference* **LoG**
 2021 *The Web Conference* **Web**
 2021 *International Joint Sunbelt and NetSci Conference* **Networks**
 2016 | 17 | 20 *European Conference on Statistical Network Science* **COSTNET**
 2017 *International Conference on Complex Networks* **CompleNet**
 2016 *International Conference on Network Science X* **NetSci-X**
 2015 | 17 *International Workshop on Social Network Analysis* **ARS**
 2015 | 16 *Networking Slovenian Networkers* **MSN**
 2013 *International World Wide Web Conference* **WWW**
 2012 *SIAM International Conference on Data Mining* **SDM**
 2012 *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* **KDD**
 2009 – 11 *European Conference on Machine Learning & [...] Knowledge Discovery [...] ECML PKDD*
 2015 | 16 *International Conference on Information Technologies and Information Society* **ITIS**
 2012 *Balkan Conference in Informatics* **BCI**
 2009 | 12 *Dnevi slovenske informatike* **DSI**
 2010 *Informatika v javni upravi* **IJU**
 2009 *Summer School on the Analysis of Patterns* **AP**

TEACHING EXPERIENCE

LECTURER

MACHINE LEARNING WITH GRAPHS

Central European University | Department of Network and Data Science
MSc and PhD students | Elective course
Sep – Dec 2025

ADVANCED TOPICS IN NETWORK SCIENCE

University of Ljubljana | Faculty of Computer and Information Science
PhD students | Elective course | With **Jure Leskovec**
Oct 2015 – Sep 2020

[INTRODUCTION TO] NETWORK ANALYSIS

University of Ljubljana | Faculty of Computer and Information Science
YouTube | <http://bit.ly/2I2dQBS>
MSc students | Mandatory course
Oct 2015 – Sep 2026

PROGRAMMING 2

University of Ljubljana | Faculty of Mathematics and Physics
YouTube | <http://bit.ly/3jRSS5B>
BSc students | Mandatory course
Oct 2018 – Sep 2026

INTRODUCTION TO NETWORK ANALYSIS IN PYTHON

University of Ljubljana | Faculty of Computer and Information Science
GitHub | <http://github.com/lovre/netpy24>
Academics and professionals | Workshop
Sep 2019 | Dec 2019 | Dec 2021 | Jan 2022 | Dec 2022 | Oct 2024

TEACHING ASSISTANT

MACHINE LEARNING WITH GRAPHS

University of Ljubljana | Faculty of Computer and Information Science
BSc and MSc students | Elective course | Lecturer **Jure Leskovec**
Oct 2013 – Sep 2026

INTRODUCTION TO INFORMATION SYSTEMS

University of Ljubljana | Faculty of Computer and Information Science
BSc students | Mandatory course | Lecturer **Dejan Lavbič**
Oct 2015 – Sep 2016

FUNDAMENTALS OF DATABASES

University of Ljubljana | Faculty of Computer and Information Science
BSc students | Mandatory course | Lecturer **Marko Bajec**
Oct 2011 – Sep 2022

DATABASES 1

University of Ljubljana | Faculty of Computer and Information Science
BSc students | Mandatory course | Lecturer **Marko Bajec**
Oct 2008 – Sep 2011

THESIS ADVISORY | BSc , MSc & PHD THESES

- **Mai Ngoc Nong** | “Bridging temporal and heterophilic graph learning for first-party financial fraud detection” | MSc thesis (Central European University, Department of Network and Data Science, 2026) | With **Petra Kralj Novak**
- **Sara Oblak** | “Enhancing dynamic scene reconstruction with deformation priors” | MSc thesis (University of Ljubljana, Faculty of Computer and Information Science & Faculty of Mathematics and Physics, 2026) | With **Matan Atzmon**
- **Gregor Mlinarič** | “Decision support systems for penicillin production optimization” | MSc thesis (University of Ljubljana, Faculty of Computer and Information Science & Faculty of Mathematics and Physics, 2025) | With **Miha Glavan** | **Society award**
- **Martin Jurkovič** | “Graph representation learning for evaluation of synthetic relational data” | MSc thesis (University of Ljubljana, Faculty of Computer and Information Science, 2025)
- **Đorđe Stevanović** | “Relational deep learning over bibliographic data” | BSc thesis (University of Ljubljana, Faculty of Computer and Information Science & Faculty of Mathematics and Physics, 2025)

- **Lan Sevčnikar** | “Automatic playlist completion using LightGCN” | BSc thesis (University of Ljubljana, Faculty of Computer and Information Science & Faculty of Mathematics and Physics, 2025)
- **Mark David Longar** | “The influence of the relational schema on the performance of graph neural networks” | BSc thesis (University of Ljubljana, Faculty of Computer and Information Science & Faculty of Mathematics and Physics, 2025)
- **Vid Stropnik** | “An analysis of player-formation graphs for predicting football passes” | MSc thesis (University of Ljubljana, Faculty of Computer and Information Science, 2024) | **Prešeren award**
- **Matej Kalc** | “Automobile fraud detection with machine learning on graphs” | MSc thesis (University of Ljubljana, Faculty of Computer and Information Science, 2024) | With **Marija Gorenc Novak**
- **Gregor Ažbe** | “Automatic invoice data extraction” | MSc thesis (University of Ljubljana, Faculty of Computer and Information Science, 2024) | With **Slavko Žitnik**
- **Mark Bogataj** | “Leakage localization in city water distribution network” | MSc thesis (University of Ljubljana, Faculty of Computer and Information Science, 2024)
- **Andraž Matkovič** | “The contribution of temporal information in the computation of brain functional connectivity when assessing its association with interpersonal differences in behaviour” | **PhD thesis** (University of Ljubljana, Faculty of Arts, 2024) | With **Grega Repovš**
- **Ermin Omeragić** | “Classification of financial transactions in accounting using graph neural networks” | MSc thesis (University of Ljubljana, Faculty of Computer and Information Science, 2023)
- **Maruša Oražem** | “Modeling product stickiness in a grocery store” | MSc thesis (University of Ljubljana, Faculty of Computer and Information Science, 2023)
- **Bisera Nikoloska** | “Network representational learning for classification of patients with neurological disorders” | BSc thesis (University of Ljubljana, Faculty of Computer and Information Science, 2023) | With **Andrej Kastrin**
- **Primož Gabrovec** | “Use of social network analysis algorithms on international projects applicants’ data” | BSc thesis (University of Ljubljana, Faculty of Computer and Information Science, 2023) | With **Zoran Bosnić**
- **Romi Koželj** | “Modeling the time evolution of seismic activity using network analysis” | MSc thesis (University of Ljubljana, Faculty of Computer and Information Science, 2022) | With **Jurij Bajc** | **Prešeren award**
- **Žiga Kokelj** | “Comparison of tainting analysis methods in Bitcoin network” | MSc thesis (University of Ljubljana, Faculty of Computer and Information Science, 2021)
- **Luka Kavčič** | “Intermediacy of nodes in networks and applications in scientometrics” | MSc thesis (University of Ljubljana, Faculty of Computer and Information Science, 2021)
- **Ilya Makarov** | “Methods of network embeddings and their applications” | **PhD thesis** (University of Ljubljana, Faculty of Computer and Information Science, 2021)
- **Iza Jerman Slabe** | “Predicting hunting success of wild animals in Primorska region” | BSc thesis (University of Ljubljana, Faculty of Computer and Information Science & Faculty of Mathematics and Physics, 2021)
- **Sara Oblak** | “Fraud detection in telecommunications using network analysis” | BSc thesis (University of Ljubljana, Faculty of Computer and Information Science & Faculty of Mathematics and Physics, 2021)
- **Tim Poštuvan** | “Adaptive graph neural network training objective for link prediction” | BSc thesis (University of Ljubljana, Faculty of Computer and Information Science & Faculty of Mathematics and Physics, 2021)
- **Šimen Ravnik** | “Sagittal pelvic orientation feature analysis for predicting degenerative diseases” | BSc thesis (University of Ljubljana, Faculty of Computer and Information Science, 2021)
- **Damir Varešanović** | “Comparison of algorithms for network community detection based on label propagation” | BSc thesis (University of Ljubljana, Faculty of Computer and Information Science, 2021)
- **Jure Repe** | “Predicting problem difficulty in bouldering” | BSc thesis (University of Ljubljana, Faculty of Computer and Information Science, 2019)
- **Simona Malenšek** | “Analysis of human symptoms-disease network” | BSc thesis (University of Ljubljana, Faculty of Computer and Information Science, 2019)
- **Žiga Babnik** | “Song similarity analysis based on lyrics, ratings and meta data” | BSc thesis (University of Ljubljana, Faculty of Computer and Information Science, 2019)
- **Nina Mrzelj** | “Link prediction in heterogenous networks by embedding nodes in the vector space” | MSc thesis (University of Ljubljana, Faculty of Computer and Information Science & Faculty of Mathematics and Physics, 2019)
- **Elizaveta Chernenko** | “Network determinants of conflict in business organization” | MSc thesis (National Research University, Faculty of Social Sciences, 2019)
- **Robert Dovžan** | “Predicting the click-through rate of ads using factorization machines” | BSc thesis (University of Ljubljana, Faculty of Computer and Information Science, 2019) | With **Davorin Kopic**
- **Luka Androjna** | “Optimization of conversions in programatic advertising” | BSc thesis (University of Ljubljana, Faculty of Computer and Information Science, 2019) | With **Andraž Tori**
- **Matic Tribušon** | “Properties of sports competitions and relation to market value” | MSc thesis (University of Ljubljana, Faculty of Computer and Information Science, 2018) | With **Erik Štrumbelj**
- **Tadej Matek** | “Anomaly detection in computer networks using higher-order dependencies” | MSc thesis (University of Ljubljana, Faculty of Computer and Information Science, 2017)
- **Petra Smole** | “Analysis of course co-enrollment at the Faculty of Computer and Information Science” | BSc thesis

(University of Ljubljana, Faculty of Computer and Information Science, 2016)

- **Rok Fortuna** | “Automated trading system using machine learning and technical analysis of trading” | BSc thesis (University of Ljubljana, Faculty of Computer and Information Science, 2016) | With **Matjaž B. Jurič**
- **Angelina Temelkovska** | “Comparison of attributes for predicting online reviews on Amazon” | BSc thesis (University of Ljubljana, Faculty of Computer and Information Science, 2016)
- **Vasil Krstev** | “Predicting performance of football teams using network analysis” | BSc thesis (University of Ljubljana, Faculty of Computer and Information Science, 2016)
- **Kristijan Mirčeta** | “Extracting temporal knowledge from events in news articles” | BSc thesis (University of Ljubljana, Faculty of Computer and Information Science, 2015)
- **Žiga Vučko** | “Predicting categories of news articles using meta-data from the Web” | BSc thesis (University of Ljubljana, Faculty of Computer and Information Science, 2015)

THESIS ASSISTANCE | BSC & PHD THESES

- **Neli Blagus** | “Effective simplification of complex real-world networks” | **PhD thesis** (University of Ljubljana, Faculty of Computer and Information Science, 2016) | Advisor **Marko Bajec**
- **Slavko Žitnik** | “Iterative semantic information extraction from unstructured text sources” | **PhD thesis** (University of Ljubljana, Faculty of Computer and Information Science, 2014) | Advisor **Marko Bajec**
- **Matjaž Hribernik** | “Predicting the outcome of badminton matches based on network analysis” | BSc thesis (University of Ljubljana, Faculty of Computer and Information Science, 2014) | Advisor **Marko Bajec**
- **Igor Jončevski** | “Predicting business ownership based on board collaboration networks” | BSc thesis (University of Ljubljana, Faculty of Computer and Information Science, 2014) | Advisor **Marko Bajec**
- **Nejc Gašperin** | “Web application for commuters” | BSc thesis (University of Ljubljana, Faculty of Computer and Information Science, 2012) | Advisor **Marko Bajec**
- **Slavko Žitnik** | “Intelligent data retrieval from arbitrary data sources” | BSc thesis (University of Ljubljana, Faculty of Computer and Information Science & Faculty of Mathematics and Physics, 2012) | Advisor **Marko Bajec**
- **Mitja Kuščer** | “Random networks generation using machine learning method” | BSc thesis (University of Ljubljana, Faculty of Computer and Information Science, 2009) | Advisor **Marko Bajec**
- **Tomaž Kuralt** | “Autonomous system for data networks integration” | BSc thesis (University of Ljubljana, Faculty of Computer and Information Science, 2009) | Advisor **Marko Bajec**

PROJECT EXPERIENCE

WISDOM LABS, D.O.O. | KNOWLEDGE GRAPHS

University of Ljubljana | Faculty of Computer and Information Science
Development of methods and algorithms for analysis of knowledge graphs
Industry funded project | Project leader
Nov 2021 – Jan 2023

COSTNET | STATISTICAL NETWORK ANALYSIS

University of Ljubljana | Faculty of Computer and Information Science
European cooperation for statistics of network data science
EU COST action | Management committee substitute
May 2016 – May 2020

ARRS | IMMIGRATION AND INTEGRATION

Faculty of Information Studies in Novo mesto
Understanding immigration and integration through social dynamics
ARRS basic project | Project member
Sep 2017 – Apr 2018

PETROL, D.D. | BUSINESS NETWORK ANALYSIS

University of Ljubljana | Faculty of Computer and Information Science
Development of methods and algorithms for network analysis in large company
Postdoctoral research project | Project leader
Jan 2014 – June 2015

CELTRA, D.O.O. | ADS ANOMALY DETECTION

University of Ljubljana | Faculty of Computer and Information Science
Development of methods and algorithms for anomaly detection in mobile ads service
Industry funded project | Project leader
Apr 2014 – Mar 2015

STREAM MINING ENVIRONMENT

University of Ljubljana | Faculty of Computer and Information Science
Development and implementation of programming environment for stream mining
Research project | Project leader
Sep 2012 — Mar 2014

ISKRATEL, D.O.O. | HIGHWAY TRAFFIC SIMULATION

University of Ljubljana | Faculty of Computer and Information Science
Development and implementation of simulation environment for highway traffic
Industry funded project | Package leader
Jan — June 2012

OPTILAB, D.O.O. | INSURANCE FRAUD DETECTION

University of Ljubljana | Faculty of Computer and Information Science
Development of methods and algorithms for fraud detection in automobile insurance
Industry funded project | Package leader
Jan 2008 — June 2010

INEA, D.O.O. | BATCH PROCESS AUTOMATION

Jožef Stefan Institute | Department of Systems and Control
Implementation of methods and algorithms for batch process automation
Industry funded project | Student work
July 2006 — June 2008

MEDICAL SIGNAL ANALYSIS

Jožef Stefan Institute | Department of Communication Systems
Implementation of methods and algorithms for signal analysis in medicine
Research project | Student work
July 2005 — Sep 2005

IT & PROGRAMMING SKILLS

General purpose languages: **PHYTON** | **JAVA** | **C/C++**
Scripting languages: **R** | **JAVASCRIPT** | **SHELL**
Query languages: **SQL** | **NOSQL** | **XQUERY**
Markup languages: **XHTML** | **XML** | **CSS**
Declarative languages: **PROLOG**
Numeric computation: **MATLAB**
Symbolic computation: **MATHEMATICA**
Digital typesetting: **TEX** | **LATEX**

LANGUAGES

SLOVENE | Native speaker
ENGLISH | Fluent
CROATIAN | Basic
GERMAN | Very basic

REFERENCES

JURE LESKOVEC | **Stanford University**, Computer Science Department
MARINKA ŽITNIK | **Harvard University**, Department of Biomedical Informatics
LUDO WALTMAN | **Leiden University**, Centre for Science and Technology Studies
VLADIMIR BATAGELJ | University of Primorska & Institute of Mathematics, Physics and Mechanics
LJUPČO TODOROVSKI | University of Ljubljana, Faculty of Mathematics and Physics
MARKO BAJEC | University of Ljubljana, Faculty of Computer and Information Science

I was born on March 7, 1986 in Ljubljana. I am a citizen of Slovenia.